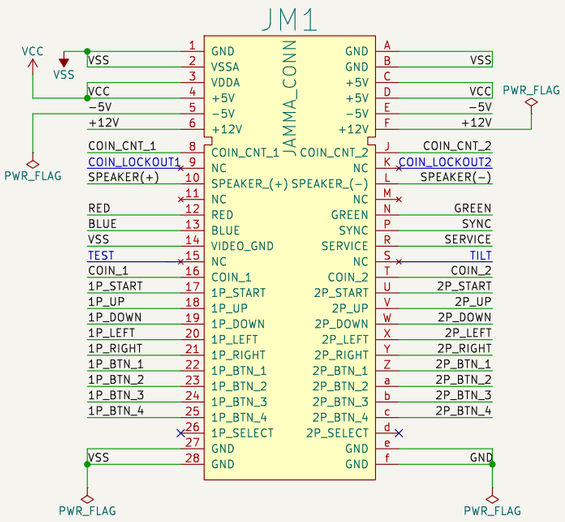
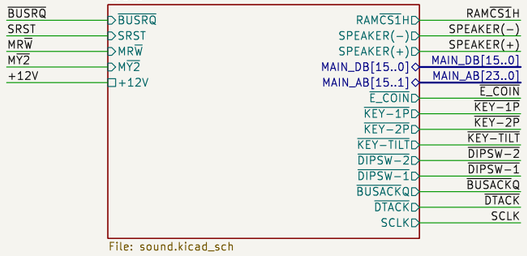
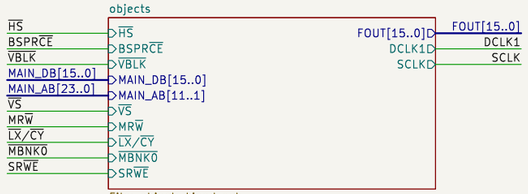
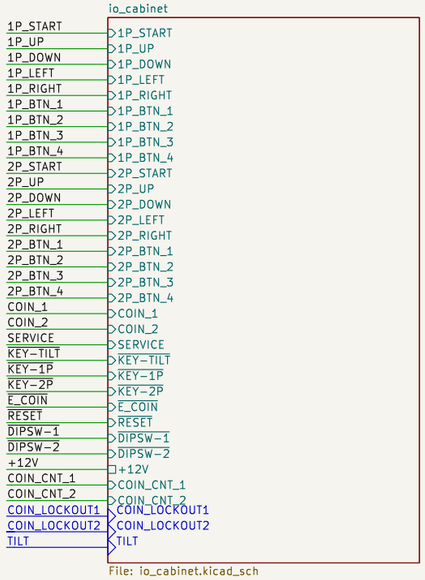
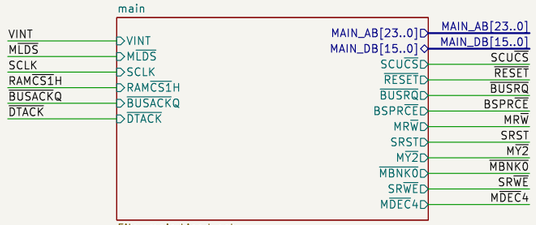










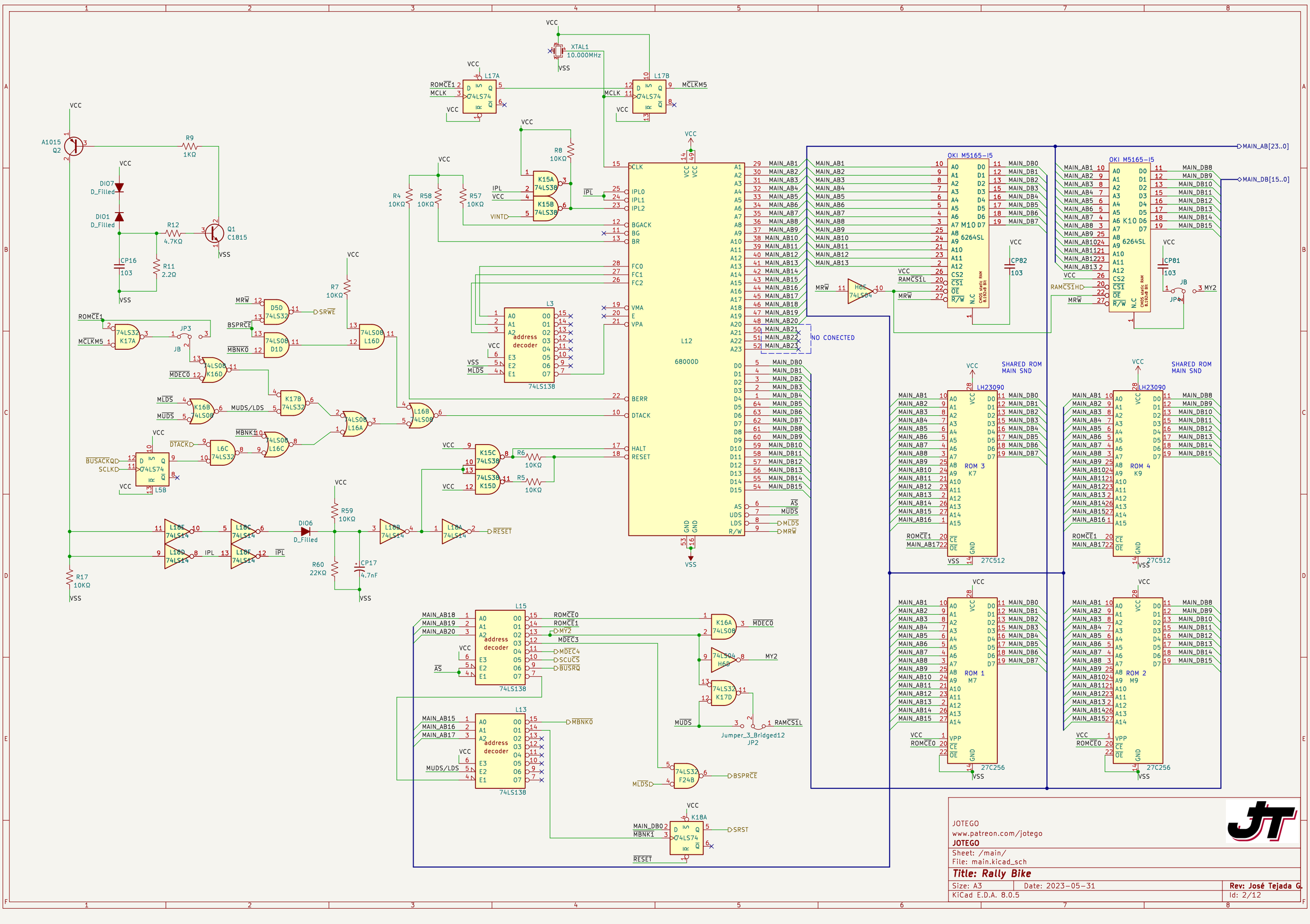


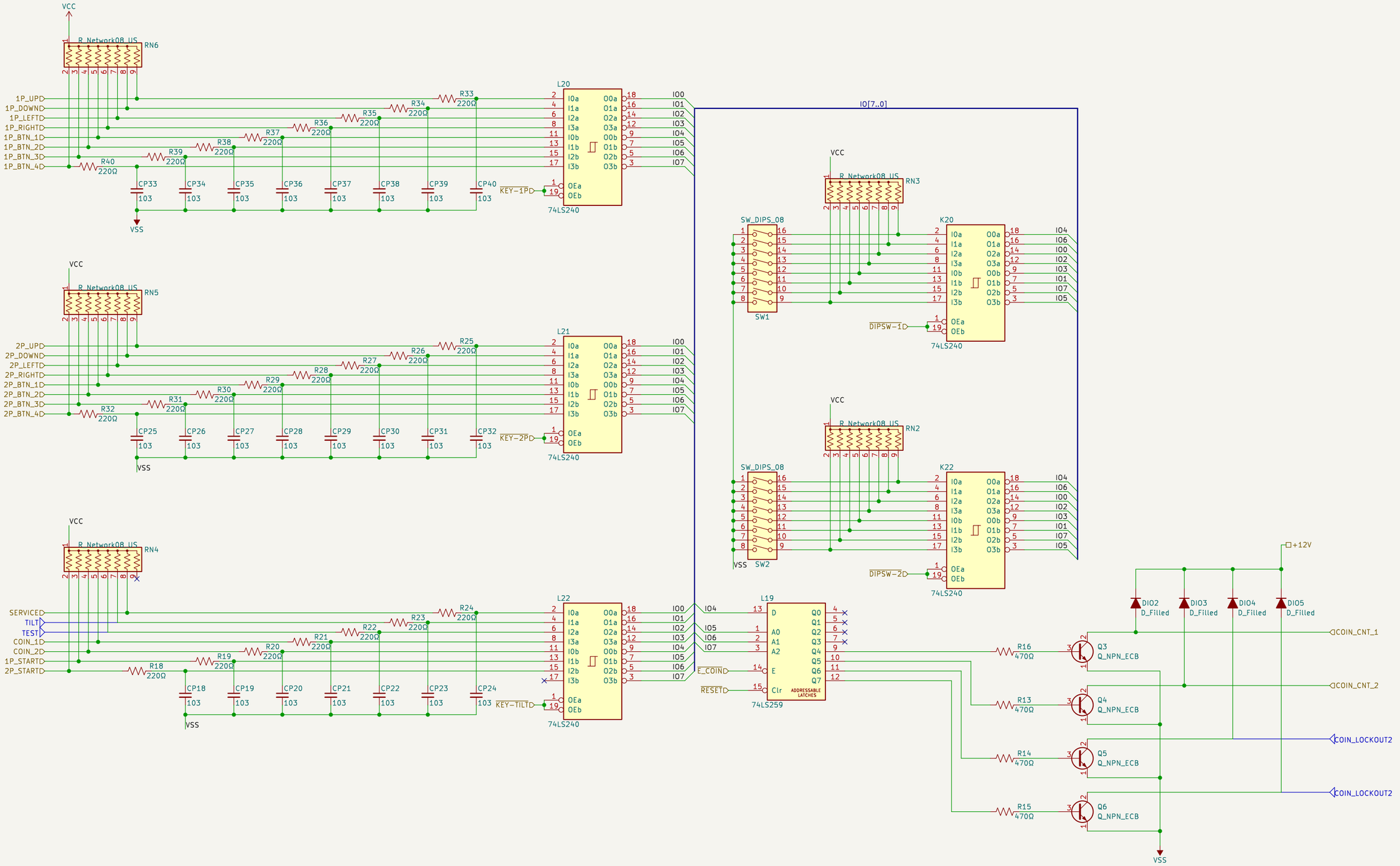
JOTEGO
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Sheet: /
File: rbike.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023−05−31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 1/12


VCC
XTAL1
10.000MHz
VSS
ROMCE1
MCLK
L17A
74LS74
MCLKM5
L17B
R8
10KΩ
K15A
74LS38
K15B
IPL
VINT
R4
R58
R57
CLK
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
MAIN_AB[23..0]
MAIN_DB[15..0]
MAIN_AB1
MAIN_AB2
MAIN_AB3
MAIN_AB4
MAIN_AB5
MAIN_AB6
MAIN_AB7
MAIN_AB8
MAIN_AB9
MAIN_AB10
MAIN_AB11
MAIN_AB12
MAIN_AB13
MAIN_AB14
MAIN_AB15
MAIN_AB16
MAIN_AB17
MAIN_AB18
MAIN_AB19
MAIN_AB20
MAIN_AB21
MAIN_AB22
MAIN_AB23
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
MAIN_DB8
MAIN_DB9
MAIN_DB10
MAIN_DB11
MAIN_DB12
MAIN_DB13
MAIN_DB14
MAIN_DB15
NO CONECTED
L12
68000D
A1015
Q2
R9
1KΩ
DIO7
D_Filled
DIO1
R12
4.7KΩ
Q1
C1815
CP16
103
R11
2.2Ω
L3
address decoder
74LS138
ROMCE1
MCLKM5
JP3
JB
MDEC0
MLDS
MUDS
MUDS/LDS
MBNK0
MRW
BSPRCE
SRWE
D5D
D1D
K16A
K16D
K16B
K17A
K17B
L16A
L16B
L16D
R7
10KΩ
BUSACKQ
SCLKQ
L5B
DTACK
L6C
MBNK1
L16C
R17
10KΩ
DIO6
R59
10KΩ
R60
22KΩ
CP17
4.7nF
L10E
L10C
L10B
L10A
L10D
L10F
RESET
R6
R5
K15C
K15D
L15
L13
MDEC3
MDEC4
SCUCS
BUSRQ
MY2
SRST
K18A
F24B
H6B
K17D
Jumper_3_Bridged12
JP2
RAMCS1L
RAMCS1H
OKI M5165-I5
6264SL
CMOS static RAM 8,192x8 Bit
CP82
CP81
JP4
SHARED ROM MAIN SND
LH23090
ROM 3
K7
ROM 4
K9
ROM 1
M7
ROM 2
M9
27C512
27C256
JOTEGO
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JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 2/12

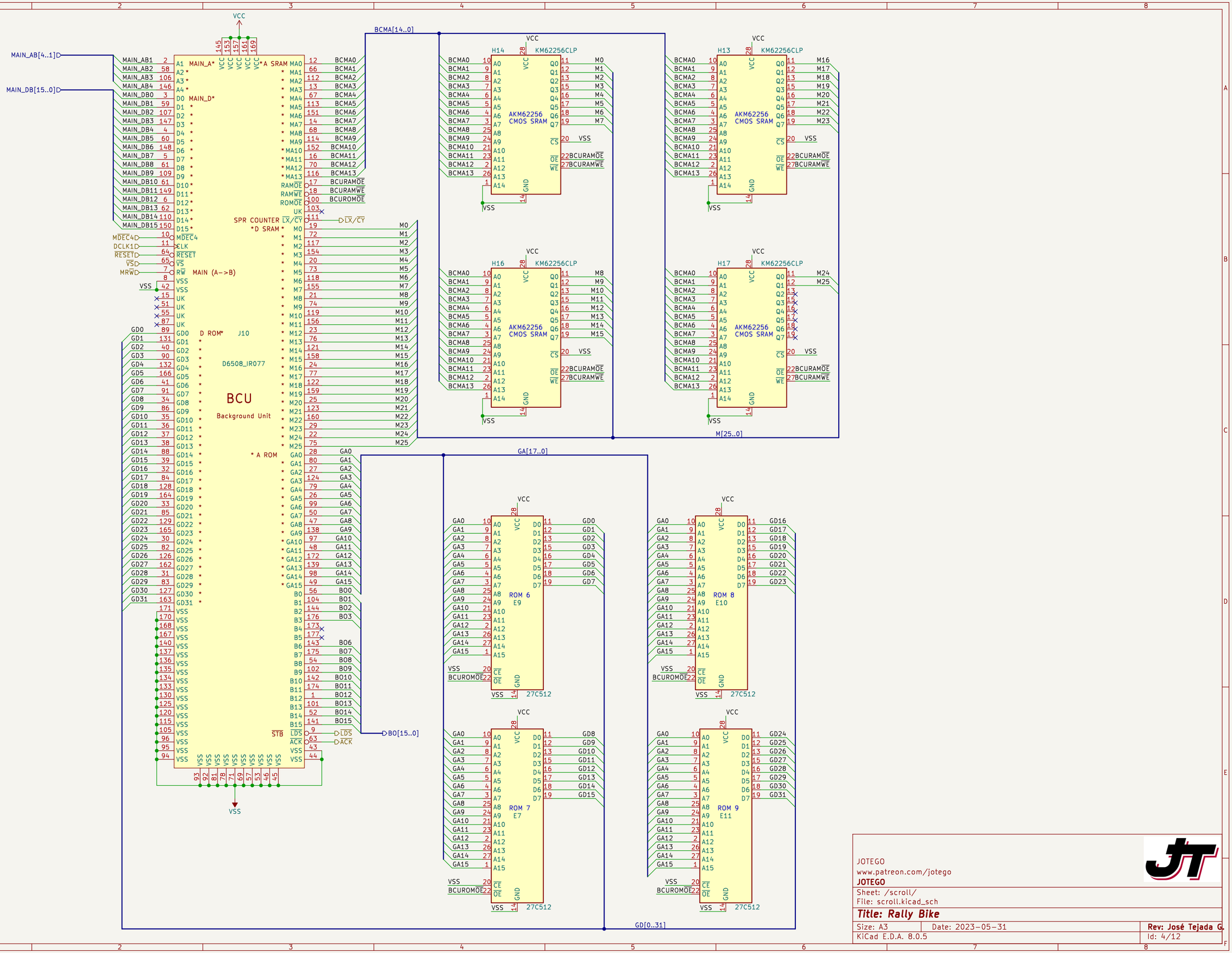


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Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 3/12


BCMA[14..0]
MAIN_AB[4..1]
MAIN_DB[15..0]
BCU
Background Unit
D6508_IR077
MAIN (A->B)
D ROM*
J10
* A ROM
*D SRAM
SPR COUNTER
BO[15..0]
GA[17..0]
GD[0..31]
M[25..0]
H14
H13
H16
H17
KM62256CLP
AKM62256
CMOS SRAM
ROM 6
E9
ROM 7
E7
ROM 8
E10
ROM 9
E11
27C512
JOTEGO
www.patreon.com/jotego
Sheet: /scroll/
File: scroll.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
KiCad E.D.A. 8.0.5
Rev: José Tejada G.
Id: 4/12

obj_memories

File: obj_memories.kicad_sch

obj_pxl

File: obj_pxl.kicad_sch

Obj_fms

File: Obj_fms.kicad_sch

COUNT DIRECTION CONTROL

DEBERIA ESTAR EN HOJA DE SCU?

SCU BUS (FRONT)

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Sheet: /objects/
File: objects.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 5/12

P POR CONSULTAR

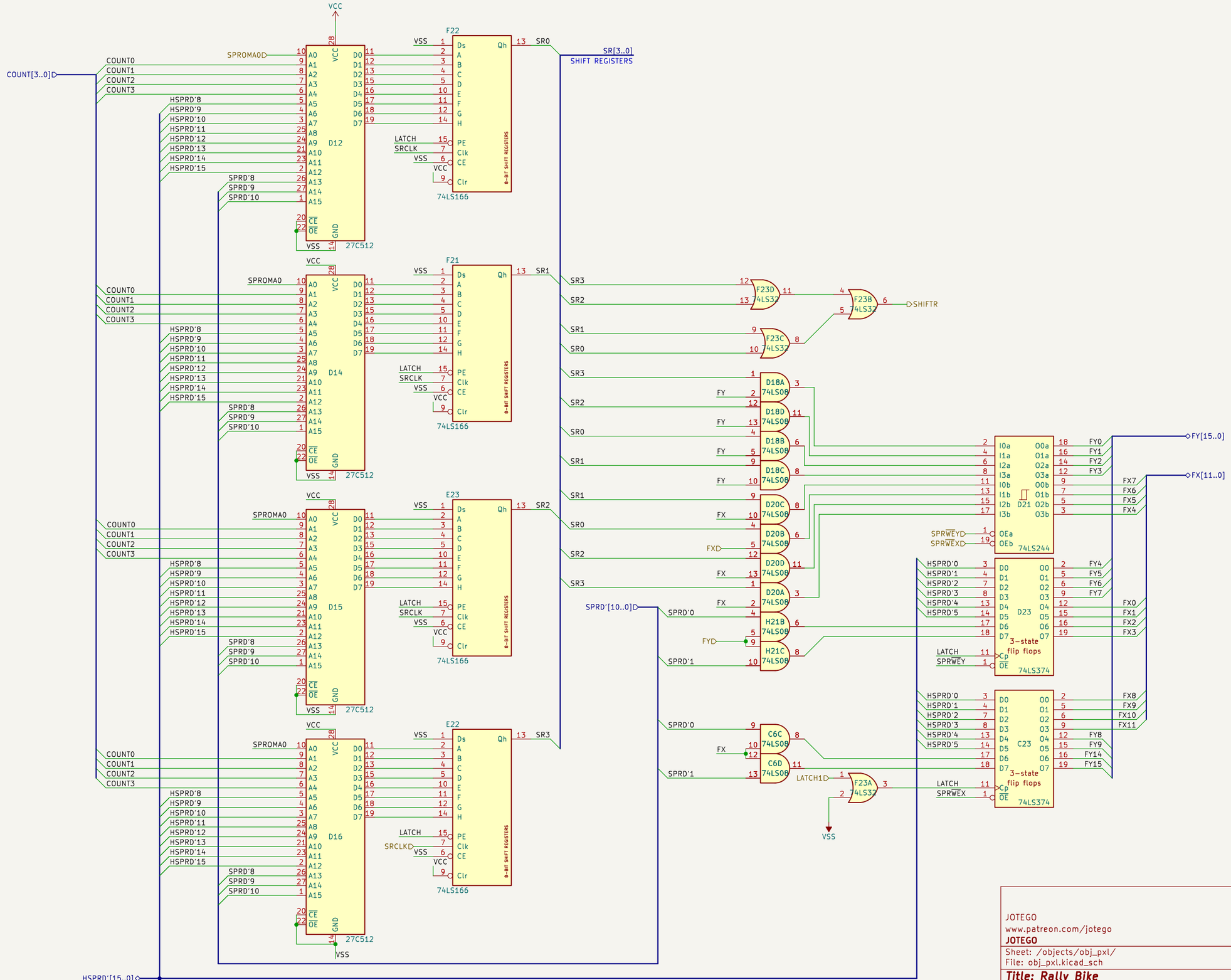
SPRITES COUNTER

SYNCHRONOUS 4-BIT UP/DOWN COUNTER

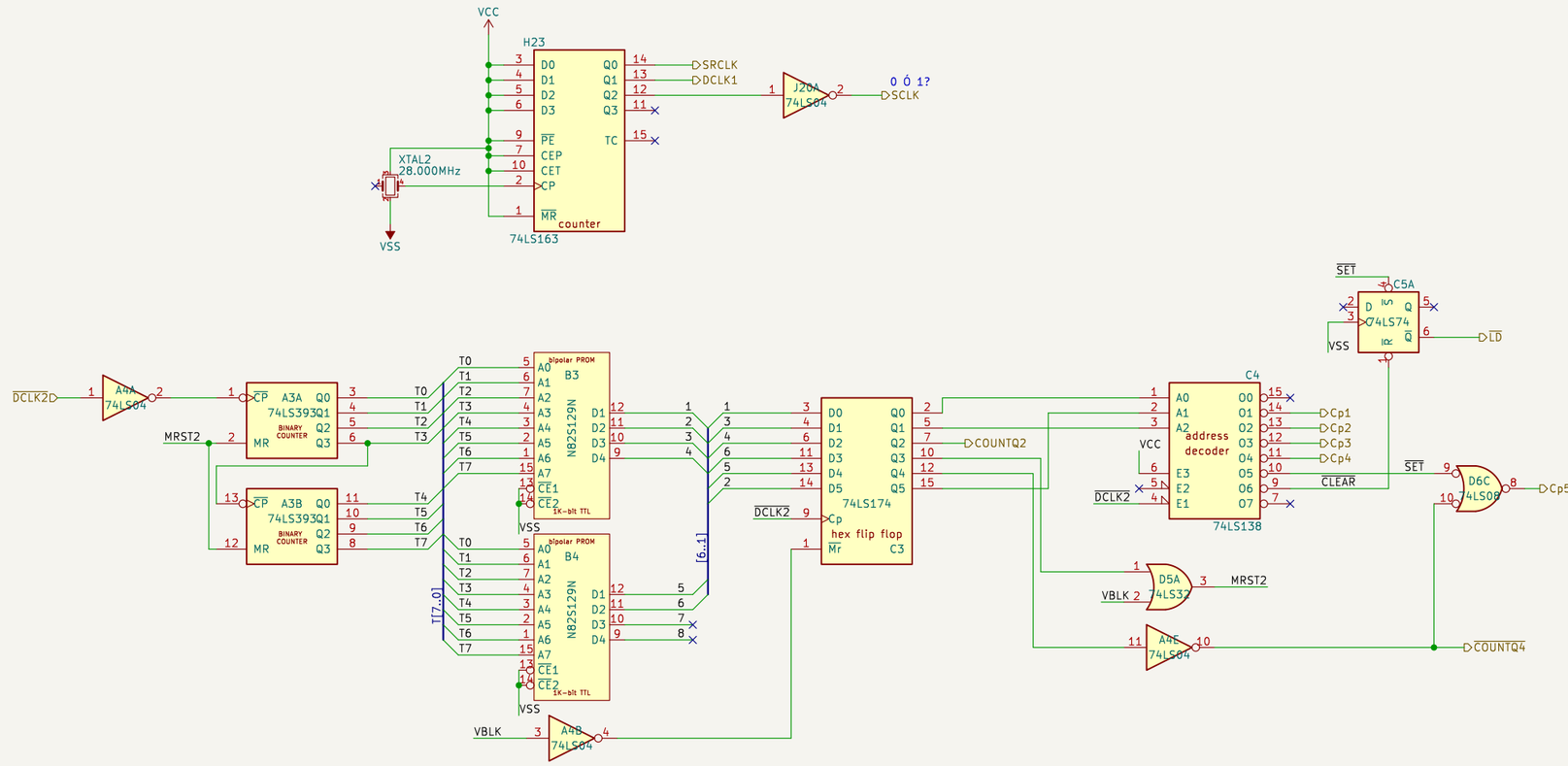
JOTEGO
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JOTEGO
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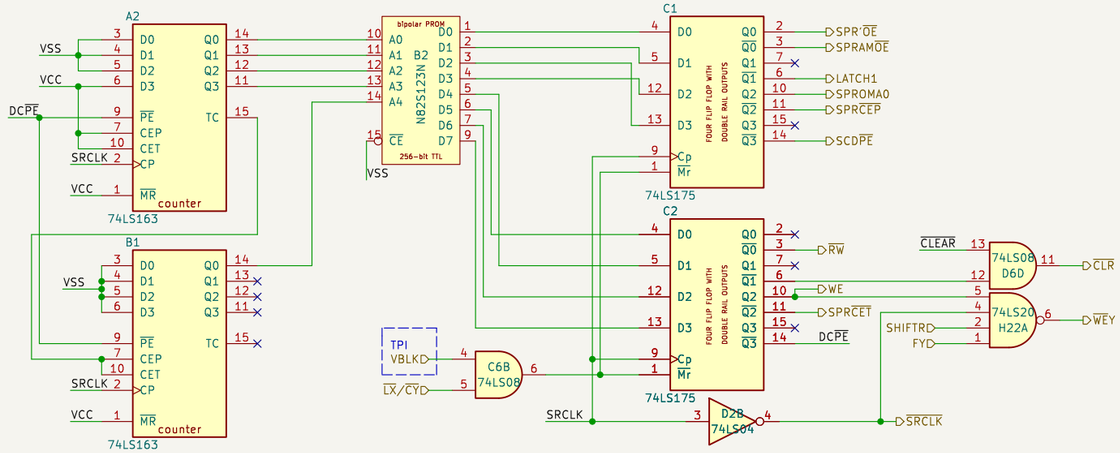
Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 6/12


COUNT[3..0]
SPROMA0
SPRD'[10..0]
HSPRD'[15..0]
SR[3..0]
SHIFT REGISTERS
SHIFTR
FY[15..0]
FX[11..0]
F22
F21
E23
E22
D12
D14
D15
D16
27C512
74LS166
8-BIT SHIFT REGISTERS
F23D
F23B
F23C
D18A
D18B
D18C
D18D
D20C
D20B
D20D
D20A
H21B
H21C
C6C
C6D
F23A
D21
D23
C23
74LS244
74LS374
3-state flip flops
JOTEGO
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JOTEGO
Sheet: /objects/obj_pxl/
File: obj_pxl.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 7/12

FINITE STATE MACHINE



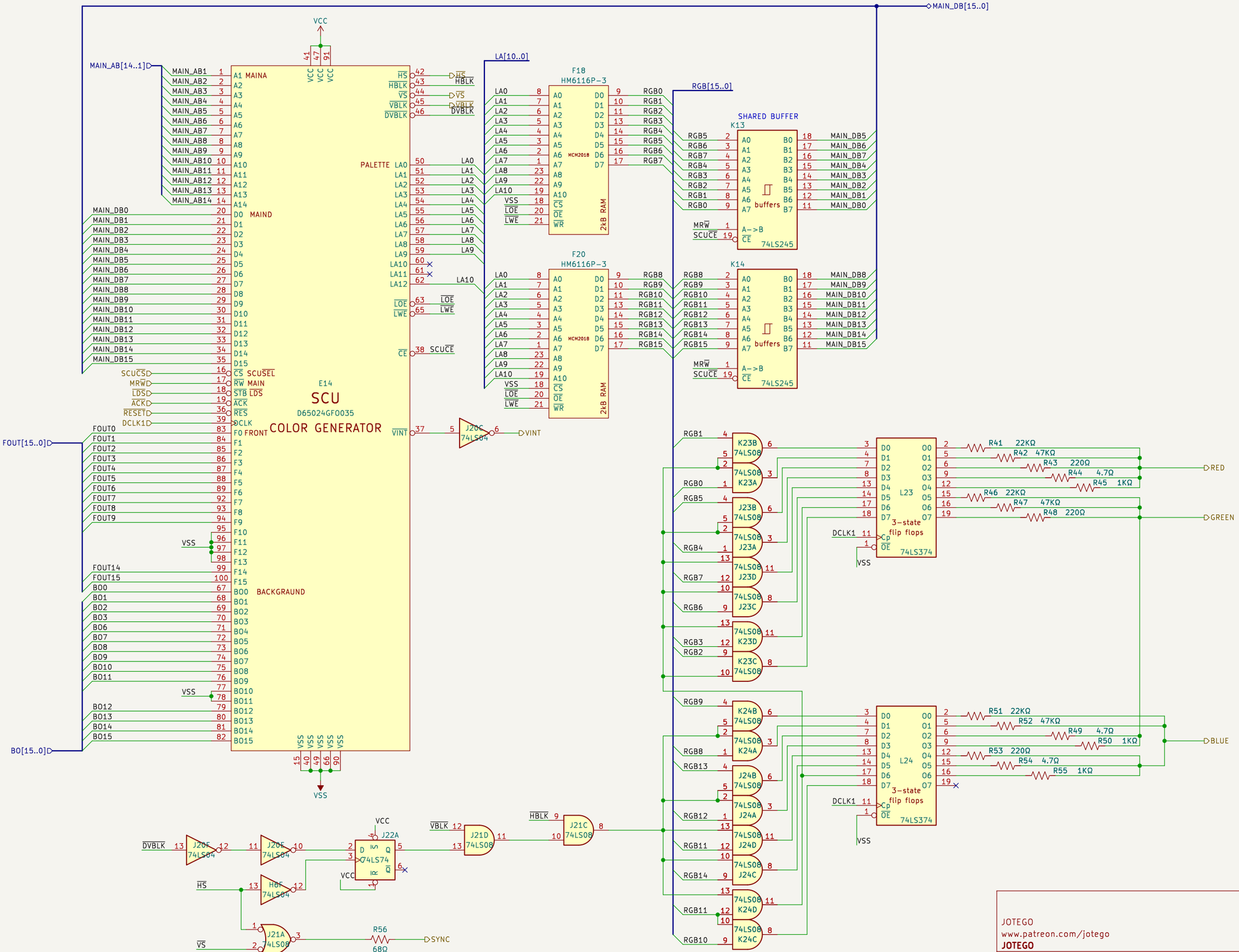


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Sheet: /objects/Obj_fms/
File: Obj_fms.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 8/12


MAIN_DB[15..0]
VCC
MAIN_AB[14..1]
LA[10..0]
RGB[15..0]
SHARED BUFFER
E14
SCU
D65024GF0035
COLOR GENERATOR
BACKGRAUND
FOUT[15..0]
BO[15..0]
F18
HM6116P-3
2kB RAM
K13
K14
74LS245
buffers
F20
L23
L24
3-state
flip ops
74LS374
RED
GREEN
BLUE
SYNC
R41 22KΩ
R42 47KΩ
R43 220Ω
R44 4.7Ω
R45 1KΩ
R46 22KΩ
R47 47KΩ
R48 220Ω
R51 22KΩ
R52 47KΩ
R49 4.7Ω
R50 1KΩ
R53 220Ω
R54 4.7Ω
R55 1KΩ
R56 68Ω
VSS
VINT
JOTEGO
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JOTEGO
Sheet: /colmix_rgb/
File: colmix_rgb.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 9/12

SHARED BUFFER'S

K1 YM3812-3526 FM OPL
J6 buffers 74LS245
M3 YM3014
M4A M4B LM358
H5 Z80CPU
H3 27128
F1 F2 K3 buffers 74LS245
H1 HM6116P-3 MCM2018 OKI M5126-25 16 BITS SRAM 2KB RAM
K6 74LS244
K5A decoder 74LS139
L14 address decoder 74LS138
M1 MB3730 INV. AMP
LS1 Speaker
Jumper_3_Bridged12
R_Network08_US

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Sheet: /sound/
File: sound.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 10/12

PWR_FLAG

VCC

CP41 103 CP42 103 CP43 103 CP44 103 CP45 103 CP46 103 CP47 103 CP48 103 CP49 103 CP50 103 CP51 103 CP52 103 CP53 103 CP54 103 CP55 103 CP56 103 CP57 103 CP58 103 CP59 103 CP60 103

CP61 103 CP62 103 CP63 103 CP64 103 CP65 103 CP66 103 CP67 103 CP68 103 CP69 103 CP70 103 CP71 103 CP72 103 CP73 103 CP74 103 CP75 103 CP76 103 CP77 103 CP78 103 CP79 103 CP80 103

CP2 10µF CP3 10µF

74LS08 K16C (10, 9, 8)

R63 470Ω CP90 10nF

J41E 74LS08 | J42E 74LS08 | L41C 74LS74 | L42E 74LS32 | H41G 74LS04 | F41E 74LS32 | E41C 74LS74 | H42C 74LS74 | J43C 74LS74 | D43E 74LS32 | J44G 74LS04 | J45E 74LS08 | H43C 74LS20 | H44E 74LS08 | L43G 74LS14

K44C 74LS139 decoder | L44C 74LS74 | L45E 74LS08

D44G 74LS04 | K45E 74LS32 | K46C 74LS74 | D41E 74LS08 | K47E 74LS08 | D42E 74LS08 | K41E 74LS08 | A41G 74LS04 | K42E 74LS08 | B5C 74LS393 BINARY COUNTER | A5C 74LS393 BINARY COUNTER | A3C 74LS393 BINARY COUNTER | D11C 74LS74 | C5C 74LS74 | D18E 74LS08 | F23E 74LS32 | C6E 74LS08

M41C LM358 (4 V− 8 V+)

E21C 74LS139 decoder

VCC

K43E 74LS38 | D7E 74LS32 | D6E 74LS08 | D1E 74LS08 | D5E 74LS32

GND

PWR_FLAG

VSS

K44B 74LS139 (A1, A0, E, O0, O1, O2, O3)

E21B 74LS139 (A1, A0, E, O0, O1, O2, O3)

B5B 74LS393 BINARY COUNTER (CP, MR, Q0, Q1, Q2, Q3)

K18B | C5B | H24B | J22B (D, CP, Q, $\overline{Q}$, $\overline{PR}$, $\overline{CL}$ 74LS74)

D1A 74LS08 | D1B 74LS08 | D1C 74LS08

D2C 74LS04 | D2B 74LS04 | D2E 74LS04 | D2F 74LS04

D7B 74LS32 | D7C 74LS32 | D7D 74LS32

D6A 74LS08 | F24A 74LS32 | K17C 74LS32

D5B 74LS32 | H6A 74LS04

J20B 74LS04 | J20D 74LS04

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File: capacitors_lu.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 11/12

Rally Bike



Rally Bike is a racing arcade video game originally developed by Toaplan and published by Taito in May 1988. In Rally Bike, players compete against computer-controlled opponents in races across locations in the United States. Initially released in arcades, the game was ported to the Nintendo Entertainment System and X68000 by different developers.
Rally Bike was met with mixed reception from video game magazines and dedicated outlets that reviewed the game when it launched in arcades and its ports. As of 2019, the rights to the title is owned by Tatsujin, a company founded in 2017 by former Toaplan member Masahiro Yuge and now-affiliate of Japanese arcade manufacturer exA-Arcadia alongside many other Toaplan IPs.

HARDWARE DESCRIPTION

Main CPU : 68000 @ 10 MHz
Sound CPU : Z80 or Z180 @ 3.5 MHz
Sound Chip : YM3812 @ 3.5 MHz
Other : Demons World has an additional TMS32010 @ 14MHz
BCU : NEC D6508IR077
SCU : NEC D65024GF035

V • T • E	Toaplan	
Games	**1980s**	*Performan* • *Tiger-Heli* • *Guardian* • *Slap Fight* • *Mahjong Sisters* • *Flying Shark* • *Wardner* • *Twin Cobra* • ***Rally Bike*** • *Truxton* • *Hellfire* • *Twin Hawk* • *Demon's World* • *Zero Wing* • *Fire Shark*
	1990s	*Snow Bros.* • *Out Zone* • *Teki Paki* • *Vimana* • *Ghox* • *Pipi & Bibi's* • *Truxton II* • *FixEight* • *Dogyuun* • *Grind Stormer* • *Knuckle Bash* • *Enma Daiō* • *Batsugun* • *Snow Bros. 2: With New Elves*
Related games	*MUSHA* • *Sorcer Striker* • *Twin Cobra II*	
People	Tatsuya Uemura • Junya Inoue • Yusuke Naora	
See also	Tamsoft • Eighting • CAVE • Gazelle • Takumi Corporation	
Category		
Portals: 1980s • Japan • Video games		

Source: https://en.wikipedia.org/wiki/Rally_Bike
http://www.system16.com/hardware.php?id=966&gid=3368#3368

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Title: Rally Bike
Size: A3 | Date: 2023-05-31 | **Rev: José Tejada G.**
KiCad E.D.A. 8.0.5 | Id: 12/12